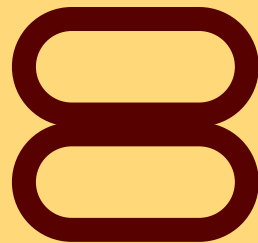
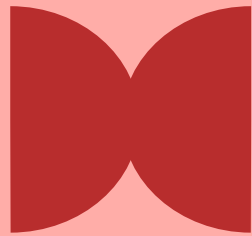


Fundamentals of Data Visualization and Communication in Dementia Research



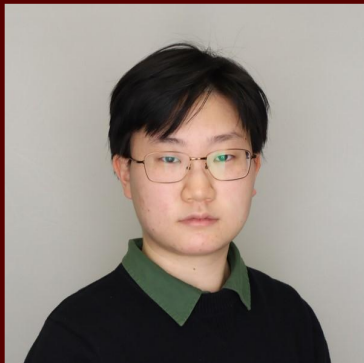
Suh Young Choi
Karen Velderrain-Lopez
Seo-Eun Choi
Shubhabrata Mukherjee

Welcome to our Immersive!

We're excited that you're here!



Facilitators



Suh Young Choi

Master's Student of Computer Science at Johns Hopkins University; part-time lecturer in the Paul G. Allen School of Computer Science & Engineering at the University of Washington



Karen
Velderrain-Lopez

Bachelor's in Computer Engineering; Research Scientist & Engineer in data science and psychometrics with the Center for Psychometric Analyses of Aging and Neurodegeneration at the University of Washington



Seo-Eun Choi

PhD in Statistics; Senior Research Scientist in psychometrics with the Center for Psychometric Analyses of Aging and Neurodegeneration at the University of Washington



Shubhabrata Mukherjee

PhD in Statistics; Research Associate Professor at the University of Washington specializing in psychometrics, statistics, and genetics as applied to Alzheimer's disease (AD) and related dementias

Immersive Agenda

1

Intro + Opening Activity

Get to know one another and
our immersive objectives

2

Visual Design Principles

What makes a good visualization?

3

Visual Accessibility

How can your visualizations
reach the widest audiences?

4

Visualization Mechanics +
Closing

What visualizations will you create?

Intros & Opening Activities

- Immersive Objectives
- Meet your tablemates!
- Color Blind Test
- Interpreting visualizations



Immersive Objectives

- Understand the basics of visual design principles and their applications in their research outputs to communicate data effectively and accurately
- Optimize designs and figures for better communication with diverse populations, including topics such as color-blindness, readability, and more
- Learn how to produce figures in R that are accessible and follow good design principles for effective communication of data and results

Intros & Opening Activities

- Immersive Objectives
- **Meet your tablemates!**
- Color Blind Test
- Interpreting visualizations



Icebreaker

Meet your tablemates!



Introduce yourself - name, pronouns

What institution are you from?
What do you do?

How have you interacted with data
visualizations? When was the last
time you made a data visualization?

What are you most excited to see in
London?

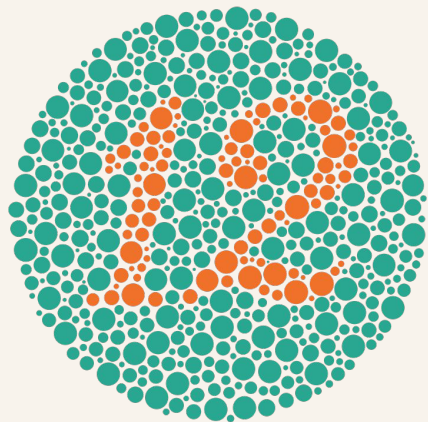
How long was your trip to London
and are there any tips for fighting jet
lag?

Intros & Opening Activities

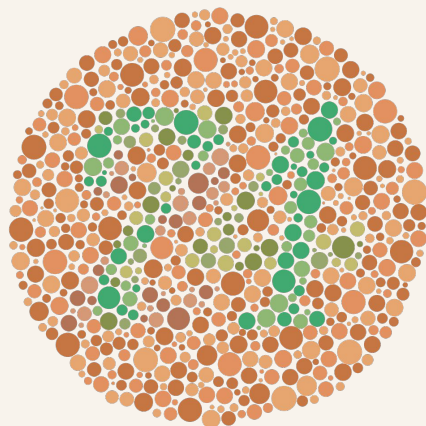
- Immersive Objectives
- Meet your tablemates!
- **Color Blind Test**
- Interpreting visualizations



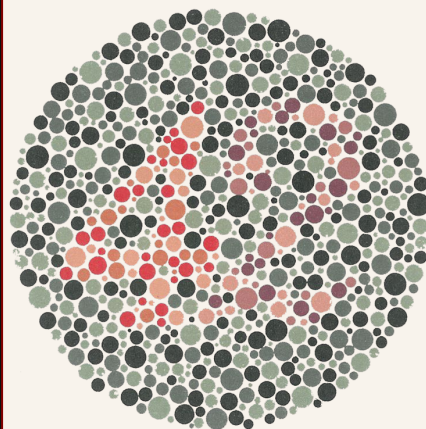
Color Blind Test



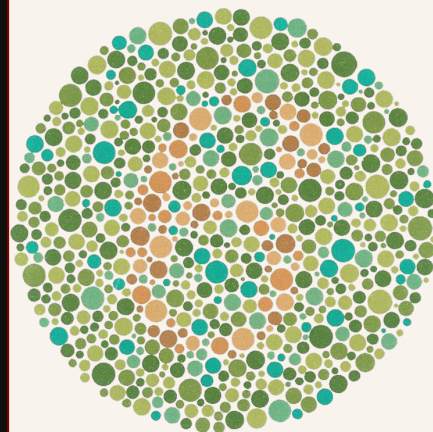
Ishihara Plate
No. 1



Ishihara Test
Plate



Ishihara Plate
No. 23



Ishihara Plate
No. 11

Intros & Opening Activities

- Immersive Objectives
- Meet your tablemates!
- Color Blind Test
- **Interpreting visualizations**

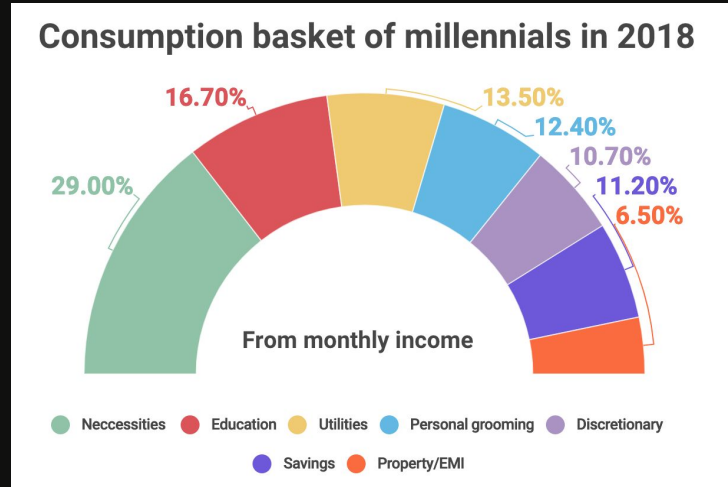


As your table goes through the visualizations, what stands out?

- Color scheme?
- Chart type?
- What is the visualization communicating? Is it doing it effectively?
- Do you like/dislike it? Why?
- What does the visualization do well?
- What does the visualization not do well?
- Any other thoughts?



Interpreting Visualizations



NSW clusters compared

Number of cases (x-axis) and days after index case when first case identified (y-axis)

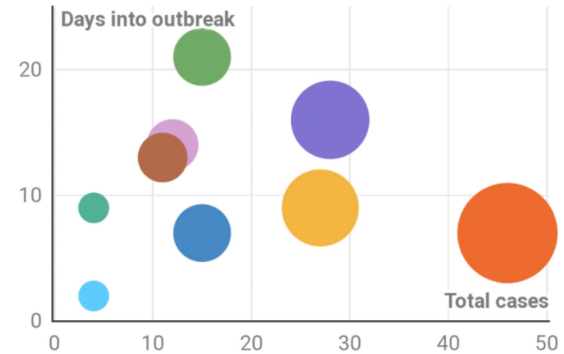
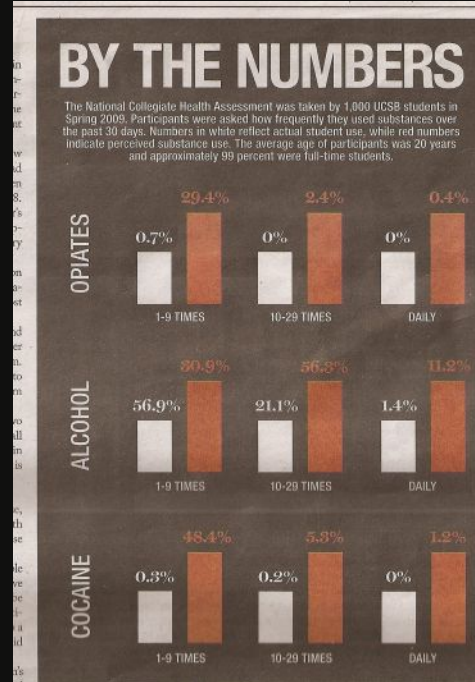
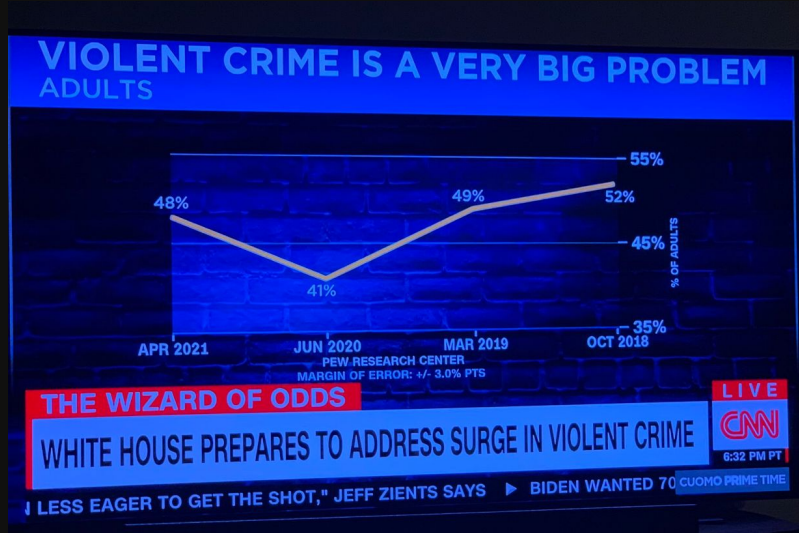
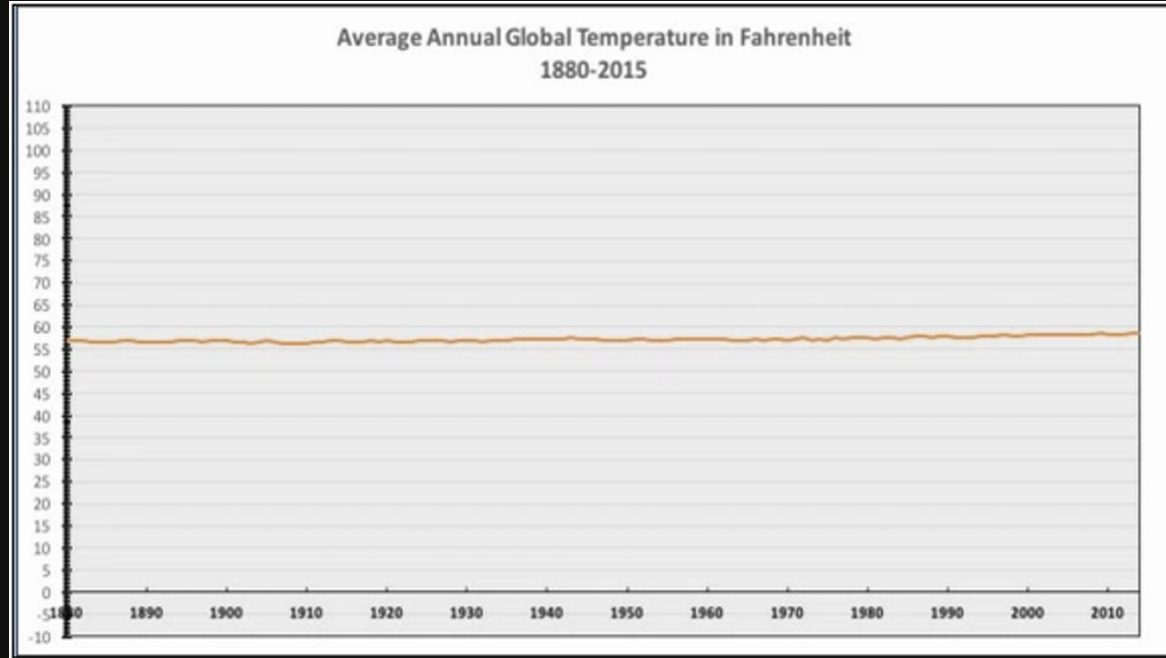


Chart: ABC News • Source: NSW Health • [Get the data](#)

Interpreting Visualizations



Interpreting Visualizations



1

2

3

4

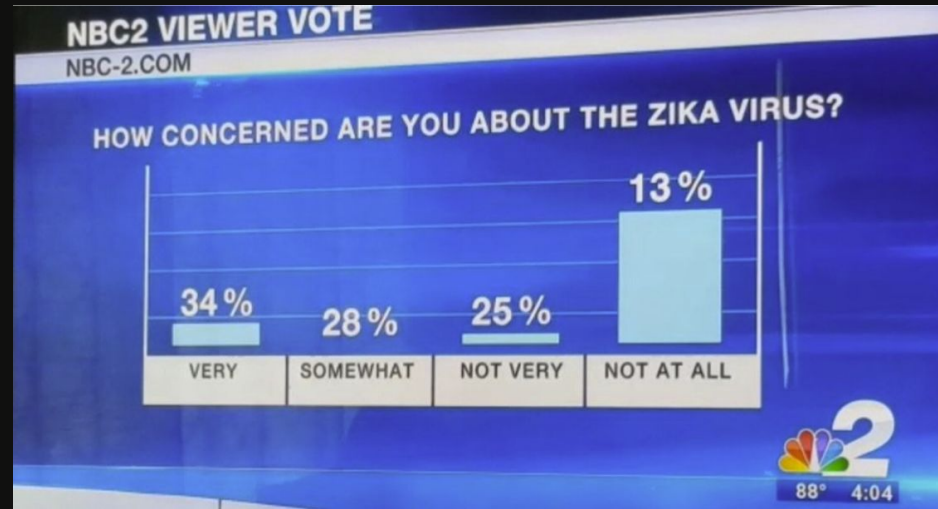
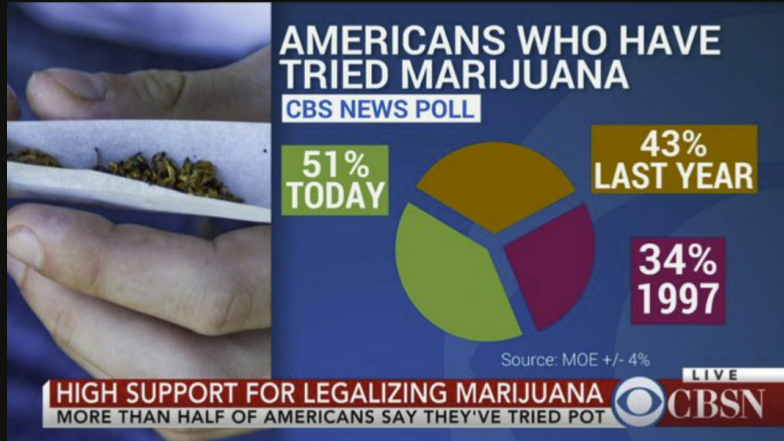
5

6

7

8

Interpreting Visualizations



1

2

3

4

5

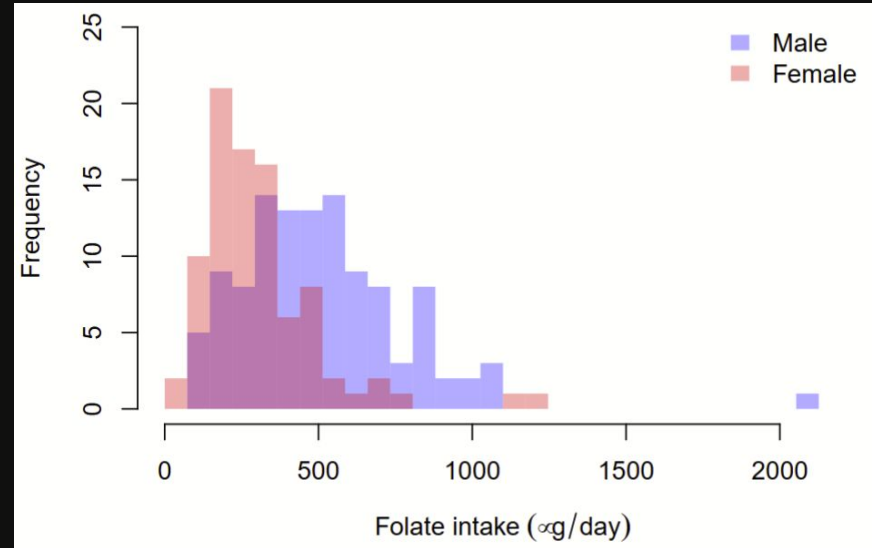
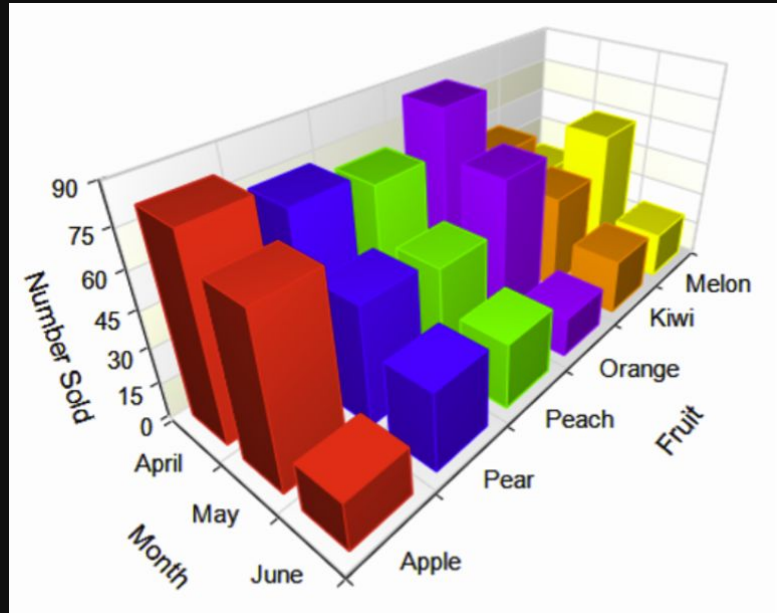
6

7

8



Interpreting Visualizations



1

2

3

4

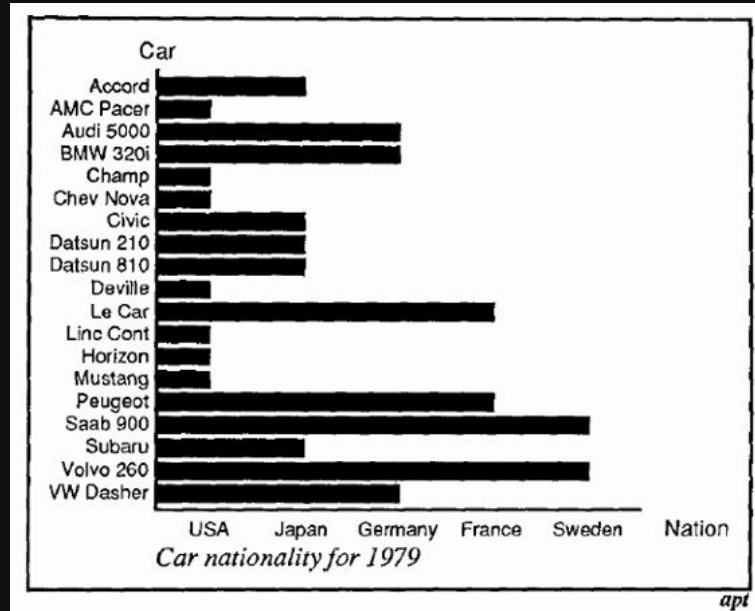
5

6

7

8

Interpreting Visualizations



1

2

3

4

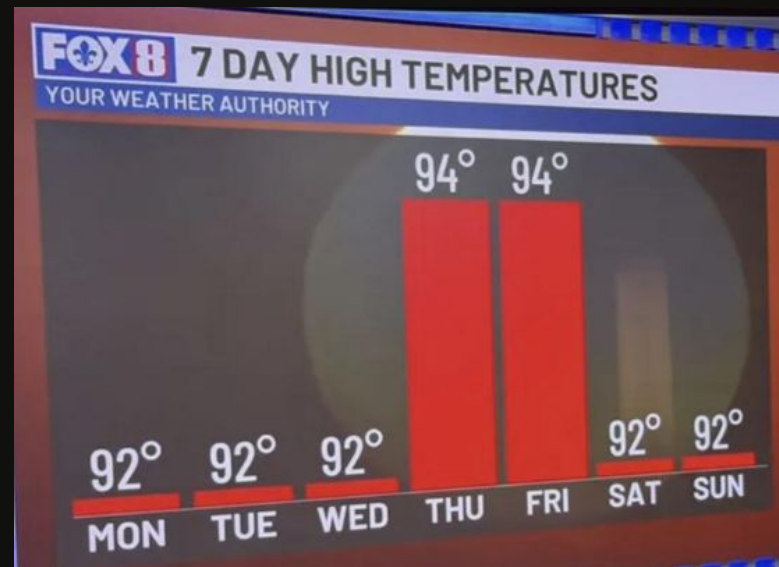
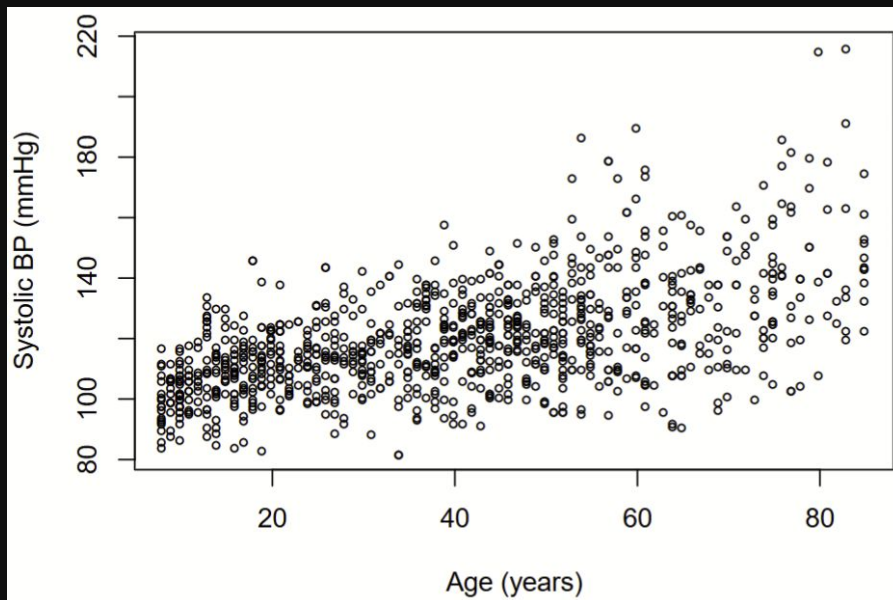
5

6

7

8

Interpreting Visualizations



1

2

3

4

5

6

7

8

As your table goes through the visualizations, what stands out?

- Color scheme?
- Chart type?
- What is the visualization communicating? Is it doing it effectively?
- Do you like/dislike it? Why?
- What does the visualization do well?
- What does the visualization not do well?
- Any other thoughts?

Now that you've gone through the visualizations, what did you think?
How can some of these visualizations be improved?

Visual Design Principles

1

What makes a good visualization?

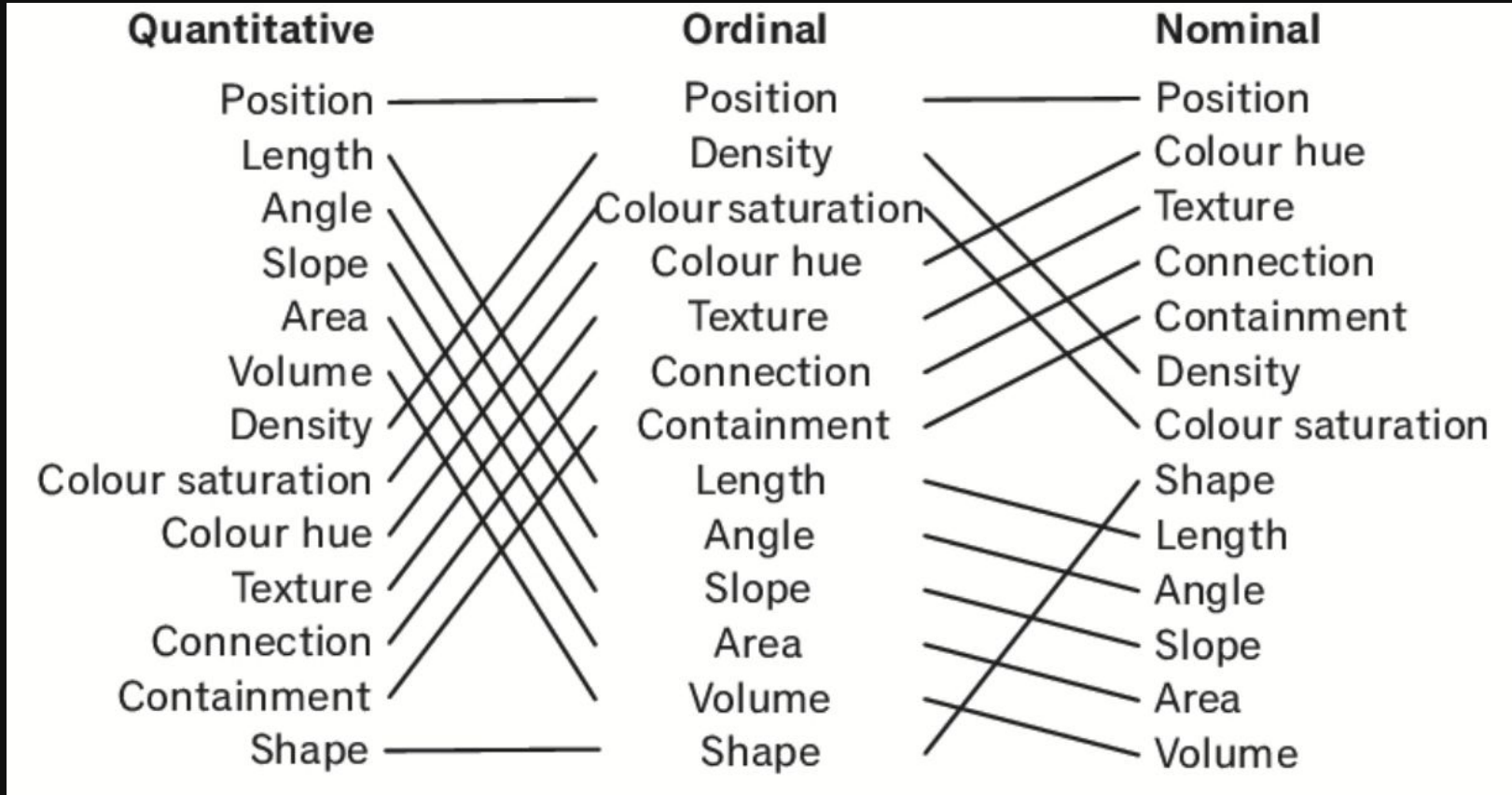
Types of Data

- Nominal
 - Categorical with no ordering
 - Fruits: apples, oranges, strawberries
 - APOE carrier status: 0, 1
- Ordinal
 - Categorical, but some ordering
 - Education: high school, associates, bachelors
 - APOE dosage: 0, 1, 2
- Quantitative
 - Numeric measurements
 - Age
 - APOE dosage: 0, 1, 2 (discrete)

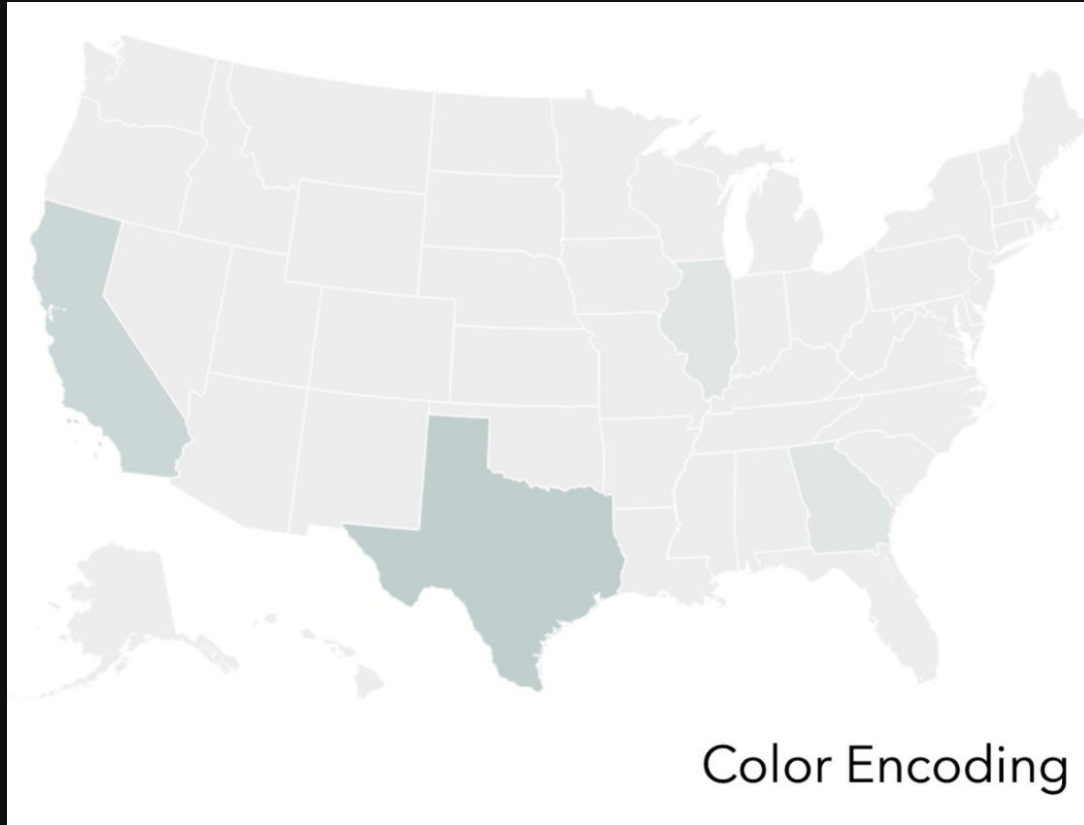
Data Encoding

- How we visually represent data
- Different encodings are more effective for different types of data
- Recall: how did the example visualizations use visual aspects to represent their data?
 - Color, size, hue

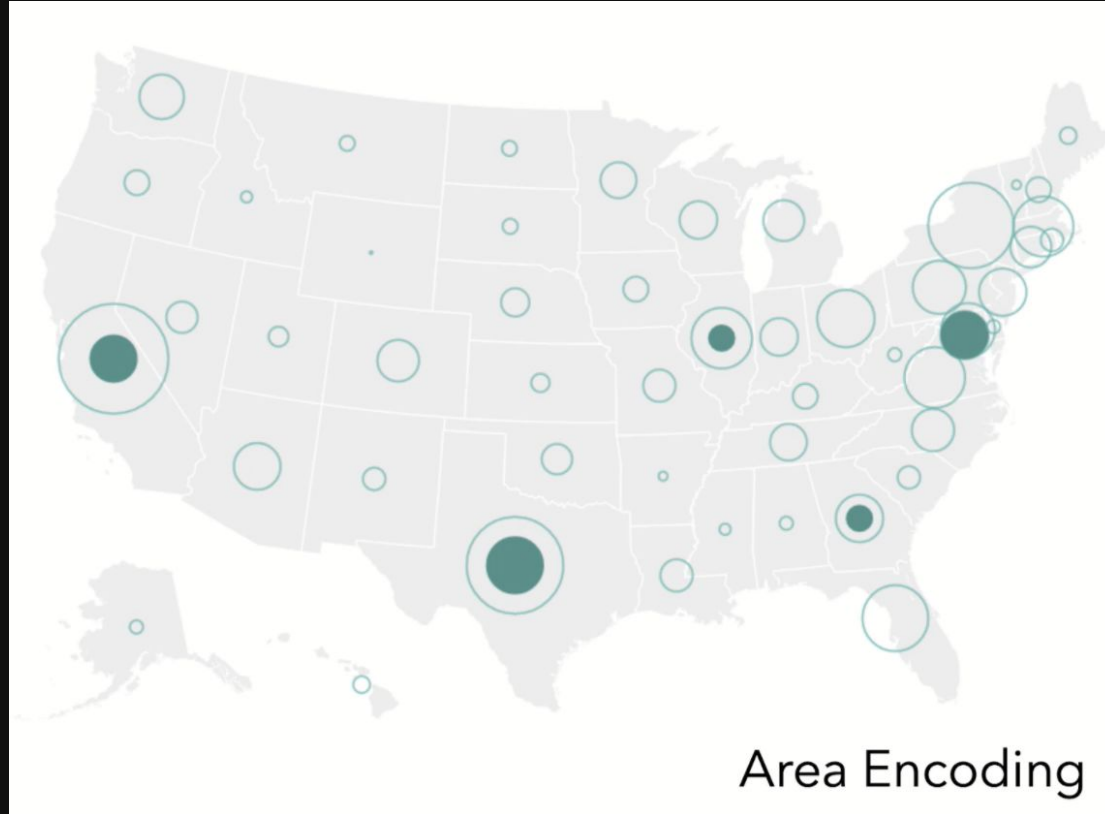
Data Encoding - Effectiveness



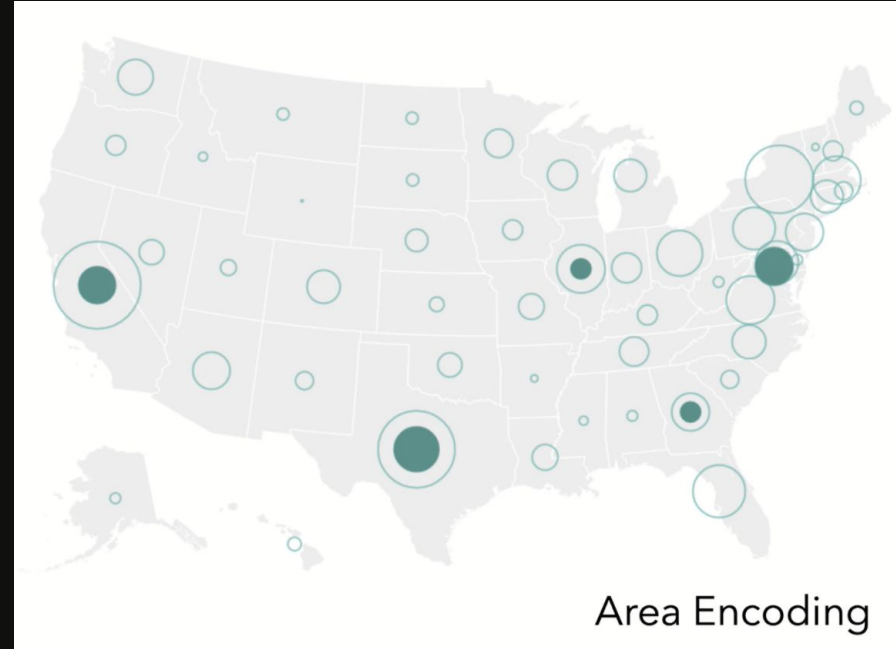
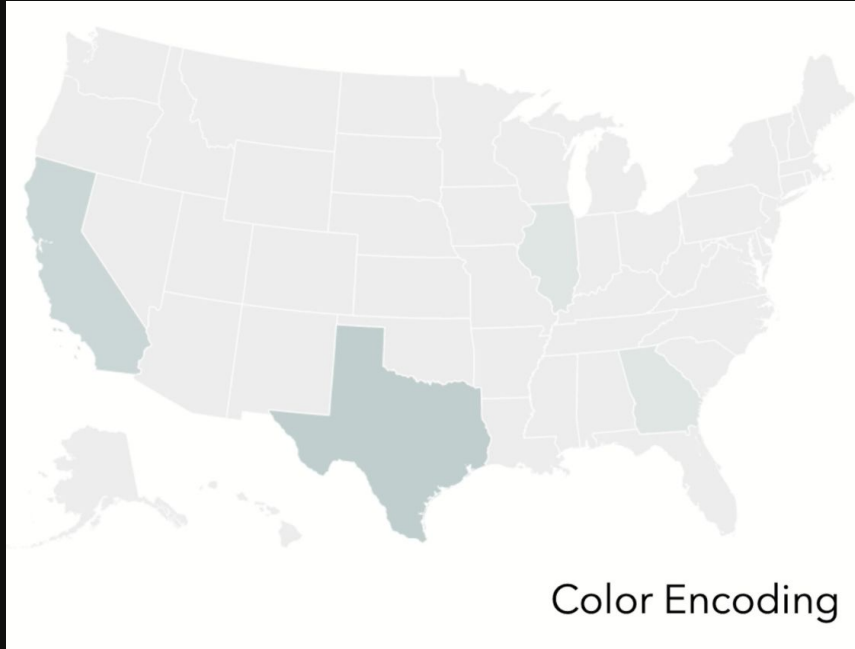
Data Encoding - Effectiveness



Data Encoding – Effectiveness



Data Encoding – Effectiveness



Data Encoding

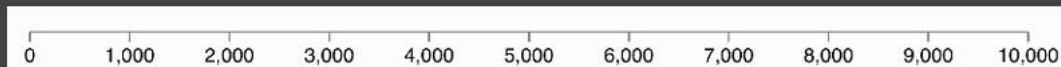
- Avoid over-encoding your data
 - Every unique data point does not need it's own unique encoding
- Use effective encodings
 - i.e. is it easier to digest the data and is it accurate
 - Encode the most important info in the most effective way
- Multi-dimensional data
 - Sometimes we want to represent many things in one data point, we can use multiple encodings together to communicate that
 - e.g. combine shape and hue

Scales & Axes

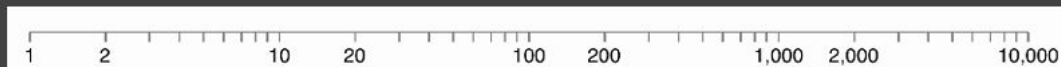
- Scales help us map data values to visual values
 - Color, x-position, angle, size
- Continuous vs discrete scales
 - Scaling the axis
- Include 0?
 - Quantitative data – interval vs ratio
- Tick marks (how many; angle; which values)

Scales & Axes

Continuous / Quantitative



linear



log

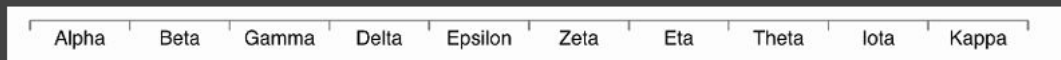


sqrt

Discrete / Ordinal



point



band

Scales & Axes

Discrete / Categorical

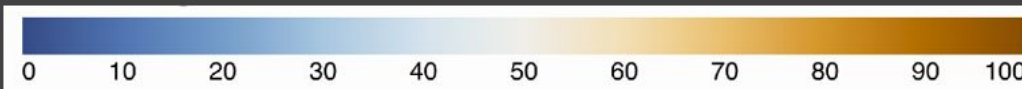


ordinal

Continuous / Quantitative

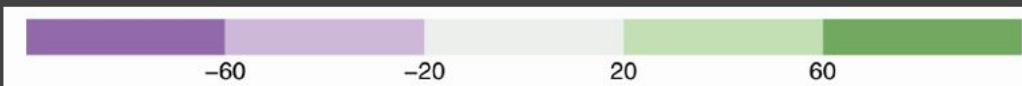


sequential



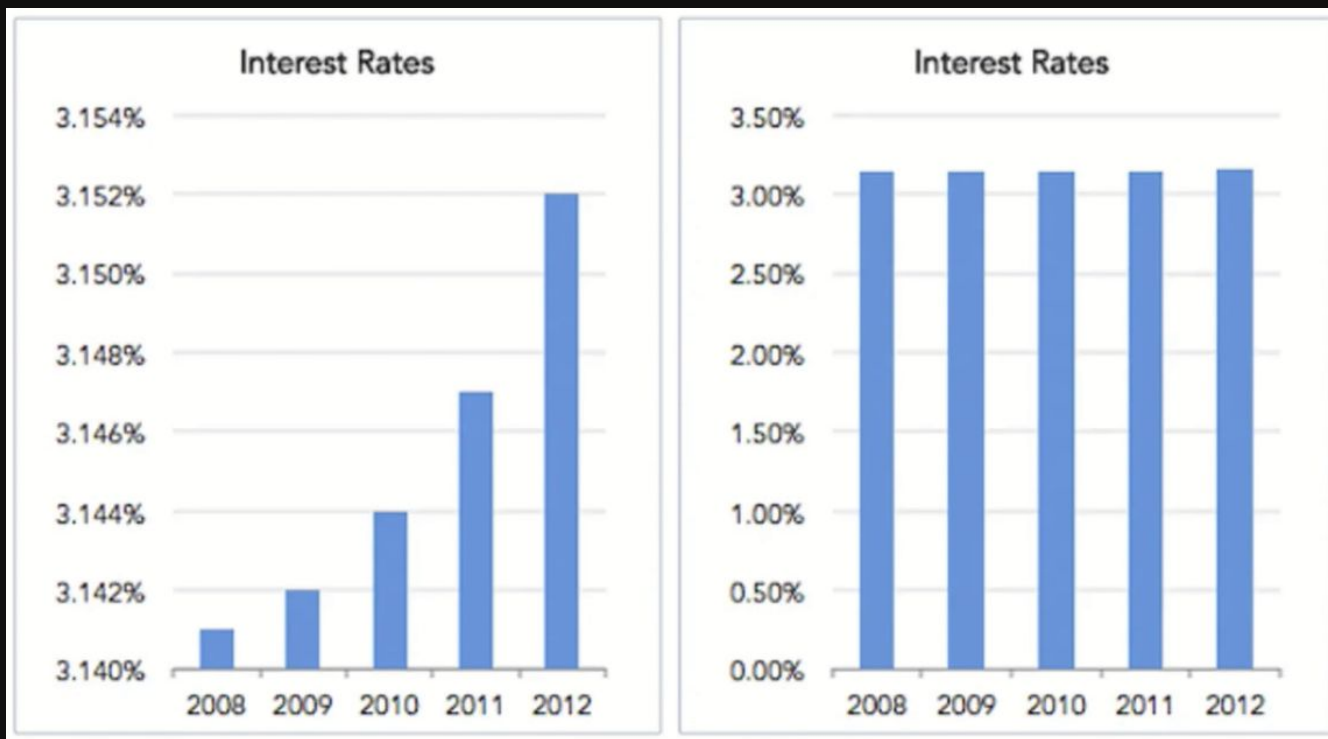
diverging

Discretized / Binned Quantitative



quantize

Scales & Axes



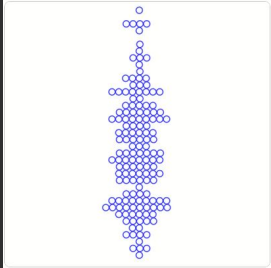
Types of Plots

- Many types of plots
- Some are more effective than others in showing data
 - Is it communicating accurately? Does it make sense?
 - What relationships are you trying to show?
 - Summary? Trends? Distribution?
- Documentation is your friend!

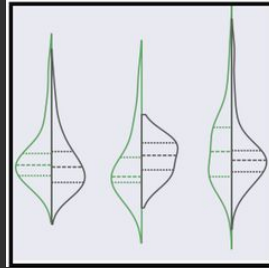
Types of Plots

Image credit: seaborn; matplotlib; r

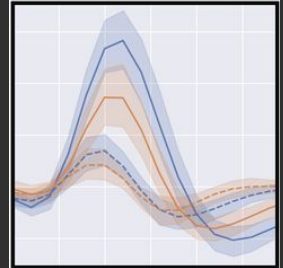
beeswarm



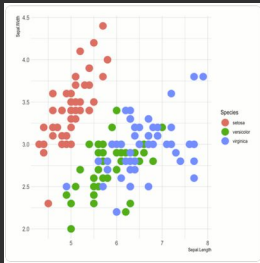
violin plot



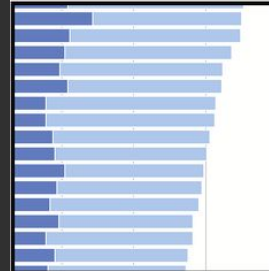
line chart



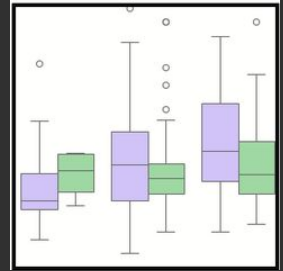
scatterplot



bar graph



boxplot



And more...

Visual Design Principles – Recap

- Know your data
- Choose a plot that communicates what you want effectively
- Use encodings that make sense and represent your data well
- Tell the truth, and only the truth with your visualizations!

Rest time

Let's take a break!



Visual Accessibility

2

How can your visualizations reach the widest audiences?

Why Accessibility Matters

- Accessibility is about care rather than compliance
- Accessible design helps more people than the group it was meant for
- Disability affects ~1 in 4 adults in the US (including your presenters)!

ACCESS IS LOVE



Color Vision Deficiency

- Affects approximately 1 in 12 men and 1 in 200 women
- Main types are protanopia/deutanopia (red/green) and tritanopia (blue-yellow)
- Try a CVD checker like Coblis!

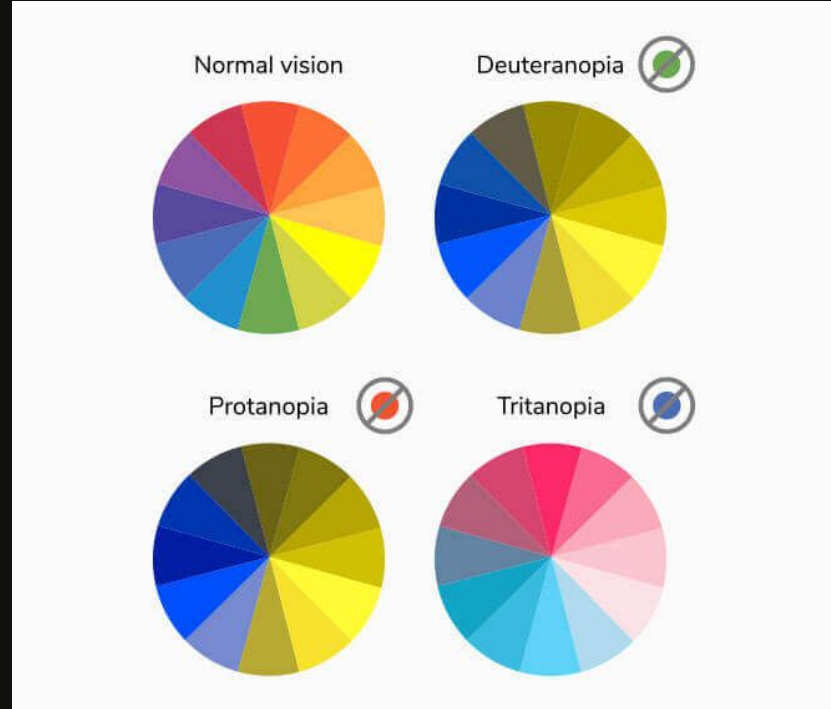


Image credit: Venngage



Contrast

- Contrast is about legibility for everyone
- WCAG contrast ratios recommend at least 4.5:1 for normal text and 3:1 for large text
- Try a contrast checker like WebAIM!

GOOD!

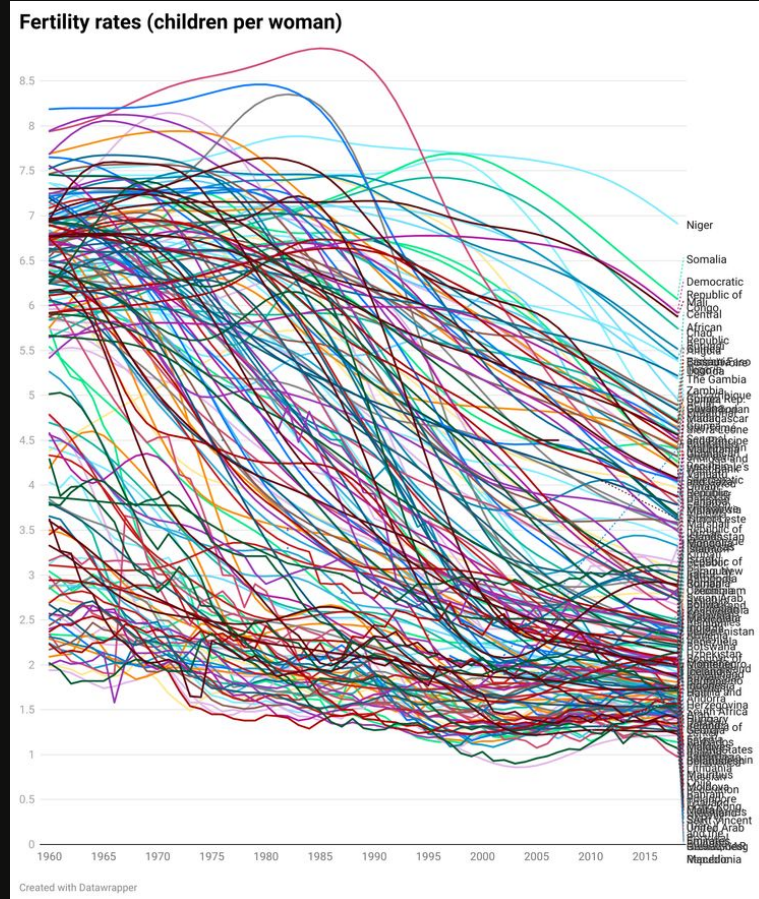
ALMOST!

YIKES!



Cognitive Load

- Clutter and crowding make visualizations difficult to read
- Use clear headings, whitespace, and grouping
- Reduce motion and unnecessary details

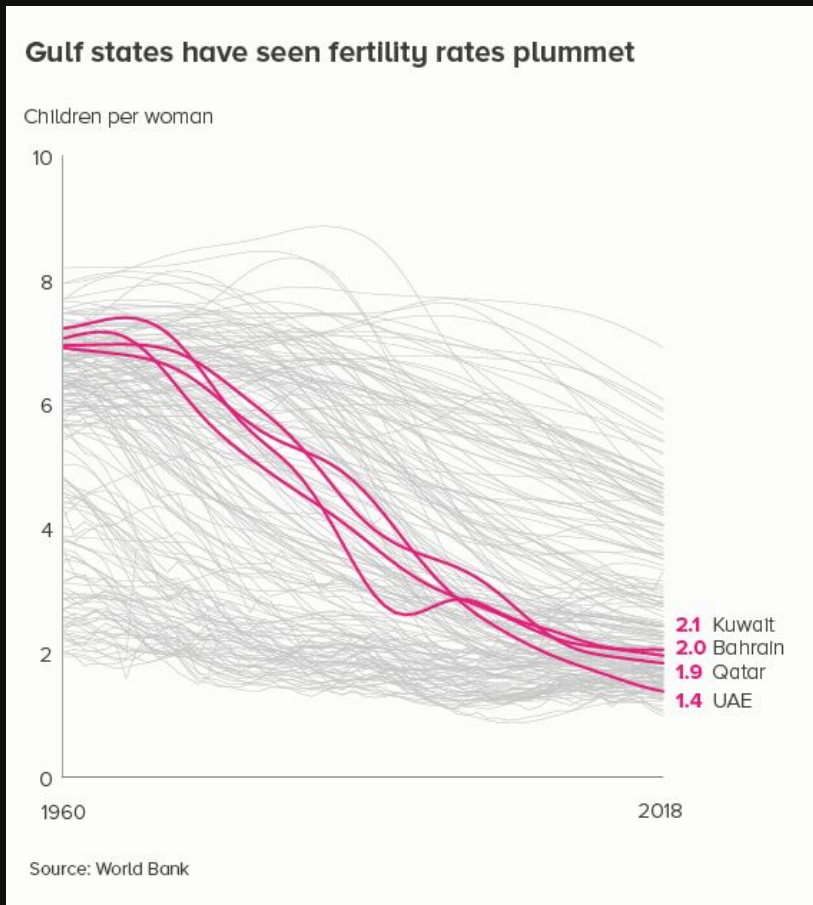


[Image sources: AddTwo Digital](#)



Cognitive Load

- Clutter and crowding make visualizations difficult to read
- Use clear headings, whitespace, and grouping
- Reduce motion and unnecessary details



[Image sources: AddTwo Digital](#)



Alt text

- Used by screen-readers and HTML
- Concise and describes purpose in addition to appearance
- Complex images require captions in addition to alt text

Alt text helps screen readers read visual content

	
Without Alt Text	With Alt Text
	
 Img_1234	 Airplane landing

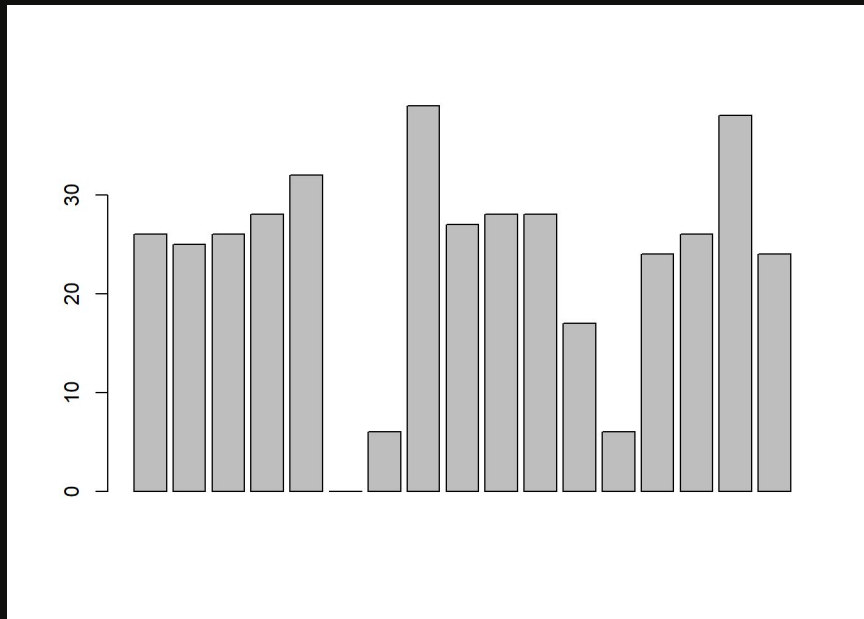
Read more: venngage.com/blog/image-alt-text

Image credit: Venngage



Labeling

- Titles, axes, and data points of interest should have labels for interpretability
- Labels should be persistent and connected to their anchor
- Consider cognitive load!



Best Practices

Never use color as the only way to convey information

Use accessibility checkers and tools often

Be consistent about layout and navigation

Include captions and labels in every visualization

Visualizations should reduce cognitive load, not add to it

Rest time

Let's take a break!



Visualization Mechanics

3

What visualizations will you create?

Coding Demo

From our Github, download and open [data_viz_accessibility_workshop.Rmd](#) from the [demo](#) folder

We'll work through these exercises together!

Takeaways

Choice of visualization matters!

Accessible design is a ground-up practice, not an afterthought!

ggplot offers many customizations for your visualizations; go see what it can do for your data!



Take a look at the Wrap-up Discussion at the end of the Rmd.

Talk with neighbors about your work, and be prepared to share with the group!





Thank you!



Further Resources

- [Harvard University Digital Accessibility Services: Data Visualizations, Charts, and Graphs](#)
- [University of Washington: Making data visualizations accessible](#)
- [data.europa.eu's Data Visualisation Guide](#)
- [ggplot2 documentation](#)

